

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Date of Assignment : 22/07/2026	Date/Time of Submission : 22/07/2026

Questions:

2a. Write a python code to create nested tuples.

Input:-

```
File Edit Format Run Options Window Help
student = ("Kaif", 23, ("BSc IT", "Semester III"), 68)

print("Name =", student[0])
print("Roll No =", student[1])
print("Course =", student[2][0])
print("Semester =", student[2][1])

print("Kaif S002")
```

Output:-

```
=====
Name = Kaif
Roll No = 23
Course = BSc IT
Semester = Semester III
Kaif S002
|
```

2b. Write a python code to sort the nested tuple using sorted () function.

Input:-

```
File Edit Format Run Options Window Help
data = [(1,2,3),(4,6,5),(9,8,7)]

sorted_data = sorted(data, key= lambda x:x[2])
print(sorted_data)

print("Kaif S002")
```

Output:-

```
=====
[(1, 2, 3), (4, 6, 5), (9, 8, 7)]
Kaif S002
|
```

2c. Write a python code to copy or clone list.

Input:-

```
File Edit Format Run Options Window Help
stud = ("Kaif",23,("BscIT","Semester III"),68)
stud2 = ()
stud2 = stud

print("Name:",stud2[0])
print("Roll no.:",stud2[1])
print("Department:",stud2[2][0])
print("Semester:",stud2[2][1])
print("Percentage:",stud2[-1])

print("Kaif S002")
```

Output:-

```
=====
Name: Kaif
Roll no.: 23
Department: BscIT
Semester: Semester III
Percentage: 68
Kaif S002
|
```

2d. Write a python code to check immutability property of python tuples.

Input:-

```
File Edit Format Run Opti
d=(6,7)
d[1]="i"

print("Kaif S002")
```

Output:-

```
Traceback (most recent call last):
  File "C:/Users/IT-22/Desktop/Jasad/PDS Practical 2(2).py", line 2, in <module>
    d[1]="i"
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80